Microstructures of SiO2 and MoS2 in Triboelectric Nanogenerators

"what kinds of microstructures have been used for SiO2 and MoS2 in TENG application? separately and the combination of these compounds'" *Generated by LeapSpace*

MoS_2 and SiO_2 have both been used in **particle, porous, nanosheet, nanoflower, fiber, and core-shell micro/nanostructured forms** in TENGs, and their **combination** is most often reported as **core-shell or encapsulated hybrid particles** that improve charge retention and humidity stability [1] [4] [11] [50] [51] [52] [70] [73].

Microstructures used separately

Material	Microstructures reported in TENGs	Typical role
SiO_2	**Nanoparticles / electrets** in PDMS; **surface microstructured films**; **porous silica-derived structures**; **amorphous SiO_2 films** from biomass-derived silica [51] [53] [54] [55]	Raises dielectric behavior, surface charge retention, and environmental stability [51] [54] [55]
MoS_2	**Monolayer/few-layer sheets**, **nanosheets in polymers**, **MoS_2@Ecoflex nanocomposites**, **crumpled/strain-induced MoS_2**, **3D porous/hierarchical MoS_2 assemblies**, and **charge-trapping layers** [1] [4] [50] [52] [56] [57] [57] [58] [61] [64] [65] [68] [69]	Acts as a charge trap, boosts tribonegative behavior, increases effective surface area, and improves output [1] [4] [52] [57] [58]

Microstructures used in the combination of SiO_2 and MoS_2

Combination structure	What was reported	Why it matters
SiO_2@MoS_2	MoS_2 nanosheets grown on SiO_2 cores / SiO_2 particles [11] [73]	MoS_2 serves as the outer tribo-active shell while SiO_2 acts as the support [11]
MoS_2@SiO_2	Silica-coated MoS_2 with hierarchical core-shell architecture [11] [70]	SiO_2 shell suppresses leakage and protects MoS_2, especially in humidity [11] [70]
MoS_2-encapsulated SiO_2 nanoparticles	Thin MoS_2 layer uniformly grown on SiO_2 [11]	Creates a hydrophobic outer layer that stabilizes charge density in humid environments [11]

Combination structure	What was reported	Why it matters
Layered / ternary stacks	Multi-shell systems containing both materials, such as **$H_2Ti_2O_5$@MoS_2@SiO_2** [71]	Shows the pair can be integrated into more complex charge-management architectures [71]

Bottom line

The main **SiO_2 microstructures** in TENGs are **nanoparticles, porous silica-based forms, and surface-microstructured films**, while **MoS_2** is most often used as **monolayers/few-layers, nanosheets, crumpled structures, and hierarchical porous assemblies** [1] [51] [53] [54] [56] [57] [58]. When combined, the dominant architecture is **core-shell or encapsulated hybrid particles**, with one material serving as the **tribo-active/charge-trapping phase** and the other as the **insulating, stabilizing, or humidity-resistant phase** [11] [70] [73].

Confidence level: **High** — the retrieved evidence repeatedly shows these same microstructure families across independent TENG studies, with the strongest recurring combination being **core-shell MoS_2/SiO_2 architectures** [11] [70] [73].

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